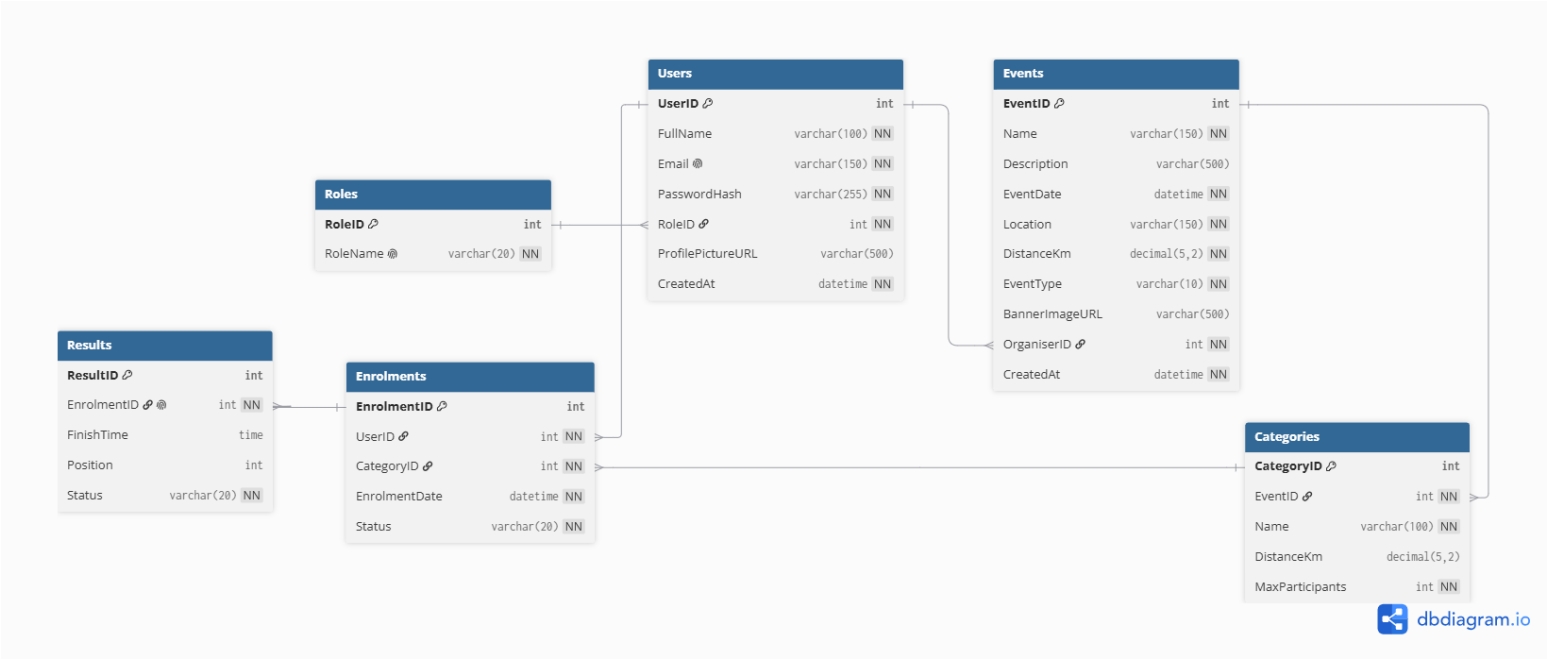
ST10493231

Tshidiso Thando Dilebo – PROG6212 – Part 1

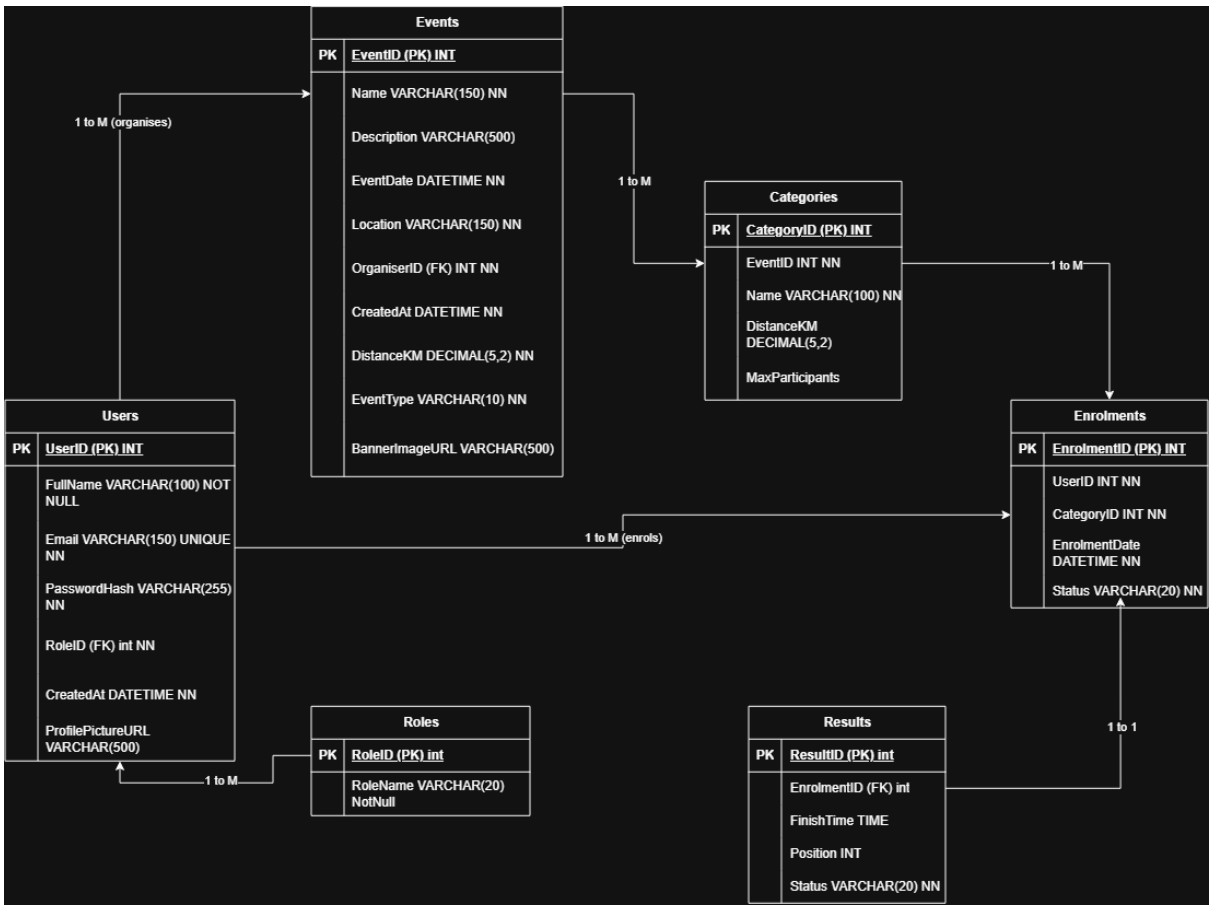
Tshidiso Thando Dilebo

09/04/2026 ST10493231

Section A



https://dbdiagram.io/d/RaceDay-ERD-6a9a9b3c5450bea1beed8bcc



Section B

RaceDay API Endpoint Plan

System**:** RaceDay: Event management platform for South African road running, walking and cycling events
Roles**:** Organiser (manages events, categories, results), Participant (browses events, enrols, views own results)

HTTP METHOD	**ROUTE**	**DESCRIPTION**	**ROLE REQUIRED**	**REQUEST BODY**	**EXPECTED RESPONCE**
POST	/api/auth/register	Creates a new user account as either an Organiser or a Participant.	None (public)	{ fullName, email, password, role }	201 Created – new user record (no password). 400 Bad Request – validation failed. 409 Conflict – email already registered.
POST	/api/auth/login	Authenticates a user and returns a JWT access token.	None (public)	{ email, password }	200 OK – { token, userId, role }. 401 Unauthorized – invalid credentials.
GET	/api/users/me	Retrieves the profile of the currently logged-in user.	Any (logged in)	None	200 OK – user profile object (includes profilePictureUrl). 401 Unauthorized – no/invalid token.
PUT	/api/users/me	Updates the profile details of the currently logged-in user.	Any (logged in)	{ fullName, email }	200 OK – updated profile. 400 Bad Request – validation failed. 401 Unauthorized.
POST	/api/users/me/profile-picture	Uploads/replaces the logged-in user's profile picture via Azure Blob Storage and stores the returned URL against their profile.	Any (logged in)	multipart/form-data – image file	200 OK – { profilePictureUrl }. 400 Bad Request – invalid file. 401 Unauthorized.

GET	/api/events	Lists all upcoming events, with optional filters (date, location).	None (public)	None	200 OK – array of events.
GET	/api/events/{id}	Retrieves full details of a single event, including its categories.	None (public)	None	200 OK – event object with categories. 404 Not Found – event does not exist.
POST	/api/events	Creates a new event. The logged-in Organiser becomes the event's owner.	Organiser	{ name, description, eventDate, location, distanceKm, eventType }	201 Created – new event record. 400 Bad Request – validation failed. 401 Unauthorized. 403 Forbidden – not an Organiser.
PUT	/api/events/{id}	Updates an existing event owned by the logged-in Organiser.	Organiser	{ name, description, eventDate, location, distanceKm, eventType }	200 OK – updated event. 403 Forbidden – not the event owner. 404 Not Found.
DELETE	/api/events/{id}	Deletes an event owned by the logged-in Organiser.	Organiser	None	204 No Content. 403 Forbidden – not the event owner. 404 Not Found.
POST	/api/events/{id}/banner	Uploads/replaces the event's banner image via Azure Blob Storage and stores the returned URL against the event.	Organiser	multipart/form-data – image file	200 OK – { bannerImageUrl }. 403 Forbidden – not the event owner. 404 Not Found.
GET	/api/events/{id}/categories	Lists all categories (e.g. 5km, 10km, 21km) for a specific event.	None (public)	None	200 OK – array of categories. 404 Not Found – event does not exist.
POST	/api/events/{id}/categories	Adds a new category to a specific event.	Organiser	{ name, distanceKm, maxParticipants }	201 Created – new category record. 403 Forbidden – not the event owner. 404 Not Found – event does not exist.

PUT	/api/categories/{id}	Updates a category's details.	Organiser	{ name, distanceKm, maxParticipants }	200 OK – updated category. 403 Forbidden. 404 Not Found.
DELETE	/api/categories/{id}	Deletes a category from an event.	Organiser	None	204 No Content. 403 Forbidden. 404 Not Found.
POST	/api/categories/{id}/enrolments	Enrols the logged-in Participant into the specified category.	Participant	{ }	201 Created – enrolment record. 404 Not Found – category does not exist. 409 Conflict – already enrolled or category full.
GET	/api/users/me/enrolments	Lists all enrolments (past and upcoming) for the logged-in Participant.	Participant	None	200 OK – array of enrolments. 401 Unauthorized.
GET	/api/events/{id}/enrolments	Lists all participants enrolled in a specific event (all categories).	Organiser	None	200 OK – array of enrolments with participant details. 403 Forbidden – not the event owner. 404 Not Found.
DELETE	/api/enrolments/{id}	Cancels an enrolment belonging to the logged-in Participant.	Participant	None	204 No Content. 403 Forbidden – not the enrolment owner. 404 Not Found.
POST	/api/enrolments/{id}/result	Captures the finish-line result for a specific enrolment.	Organiser	{ finishTime, position, status }	201 Created – result record. 403 Forbidden – not the event owner. 404 Not Found – enrolment does not exist. 409 Conflict – result already captured.
GET	/api/enrolments/{id}/result	Retrieves the result for a specific enrolment.	Any (logged in)	None	200 OK – result object. 404 Not Found – no result captured yet.

GET	/api/users/me/results	Retrieves the personal performance history of the logged-in Participant across all events.	Participant	None	200 OK – array of results with event/category context. 401 Unauthorized.

Section C

```
-- RaceDay Database
-- System Planning and Database - Part 1

IF DB_ID('RaceDayDB') IS NULL
BEGIN
  CREATE DATABASE RaceDayDB;
END

USE RaceDayDB;

-- Roles table
CREATE TABLE Roles (
  RoleID     INT IDENTITY(1,1) PRIMARY KEY,
  RoleName   VARCHAR(20) NOT NULL UNIQUE
);

-- Users table
CREATE TABLE Users (
  UserID        INT IDENTITY(1,1) PRIMARY KEY,
  FullName      VARCHAR(100) NOT NULL,
  Email         VARCHAR(150) NOT NULL UNIQUE,
  PasswordHash  VARCHAR(255) NOT NULL,
  RoleID        INT NOT NULL,
  ProfilePictureURL VARCHAR(500) NULL,
  CreatedAt     DATETIME NOT NULL DEFAULT GETDATE(),
  CONSTRAINT FK_Users_Roles FOREIGN KEY (RoleID) REFERENCES Roles(RoleID)
);

-- Events table
-- Created and owned by an Organiser (Users)
CREATE TABLE Events (
  EventID       INT IDENTITY(1,1) PRIMARY KEY,
  Name          VARCHAR(150) NOT NULL,
  Description   VARCHAR(500) NULL,
  EventDate     DATETIME NOT NULL,
  Location      VARCHAR(150) NOT NULL,
  DistanceKm    DECIMAL(5,2) NOT NULL,
  EventType     VARCHAR(10) NOT NULL,
```

```
  BannerImageURL  VARCHAR(500) NULL,
  OrganiserID    INT NOT NULL,
  CreatedAt      DATETIME NOT NULL DEFAULT GETDATE(),
  CONSTRAINT FK_Events_Users FOREIGN KEY (OrganiserID) REFERENCES
Users(UserID),
  CONSTRAINT CK_Events_EventType CHECK (EventType IN ('Run', 'Walk', 'Cycle'))
);

-- Categories table
-- Distance/race categories that belong to an Event
-- 5km, 10km, 21km
CREATE TABLE Categories (
  CategoryID     INT IDENTITY(1,1) PRIMARY KEY,
  EventID        INT NOT NULL,
  Name           VARCHAR(100) NOT NULL,
  DistanceKm     DECIMAL(5,2) NULL, -- NULL for age-based categories ('Under 20',
'Senior')
  MaxParticipants INT NOT NULL DEFAULT 100,
  CONSTRAINT FK_Categories_Events FOREIGN KEY (EventID) REFERENCES
Events(EventID)
);

-- Enrolments table
-- Junction table: a Participant (Users) enrolling in a Category.
-- Implements the many-to-many relationship between Users and Categories.
CREATE TABLE Enrolments (
  EnrolmentID    INT IDENTITY(1,1) PRIMARY KEY,
  UserID         INT NOT NULL,
  CategoryID     INT NOT NULL,
  EnrolmentDate  DATETIME NOT NULL DEFAULT GETDATE(),
  Status         VARCHAR(20) NOT NULL DEFAULT 'Confirmed',
  CONSTRAINT FK_Enrolments_Users FOREIGN KEY (UserID) REFERENCES
dbo.Users(UserID),
  CONSTRAINT FK_Enrolments_Categories FOREIGN KEY (CategoryID) REFERENCES
Categories(CategoryID),
  CONSTRAINT UQ_Enrolments_User_Category UNIQUE (UserID, CategoryID)
);

-- Results table
-- One result per Enrolment (1-to-1), captured by the Organiser after the race
CREATE TABLE Results (
```

```
  ResultID      INT IDENTITY(1,1) PRIMARY KEY,
  EnrolmentID   INT NOT NULL UNIQUE,
  FinishTime    TIME NULL,
  Position      INT NULL,
  Status        VARCHAR(20) NOT NULL DEFAULT 'Finished',
  CONSTRAINT FK_Results_Enrolments FOREIGN KEY (EnrolmentID) REFERENCES
Enrolments(EnrolmentID)
);

-- Roles
INSERT INTO Roles (RoleName) VALUES ('Organiser'), ('Participant');

-- Users
INSERT INTO Users (FullName, Email, PasswordHash, RoleID, ProfilePictureURL)
VALUES
  ('Thabo Nkosi',   'thabo.nkosi@raceday.co.za',   'hashed_pw_1', 1, NULL), -- Organiser
  ('Lindiwe Dube',   'lindiwe.dube@raceday.co.za',   'hashed_pw_2', 1, NULL), --
Organiser
  ('Sipho Mokoena',   'sipho.mokoena@example.com',   'hashed_pw_3', 2, NULL), --
Participant
  ('Anika van Wyk',   'anika.vanwyk@example.com',    'hashed_pw_4', 2, NULL), --
Participant
  ('Naledi Molefe',   'naledi.molefe@example.com',   'hashed_pw_5', 2, NULL); --
Participant

-- Events
INSERT INTO Events (Name, Description, EventDate, Location, DistanceKm, EventType,
BannerImageURL, OrganiserID)
VALUES
  ('Johannesburg City Run',  'Annual road running event through the Johannesburg CBD.',
'2026-11-14 07:00:00', 'Johannesburg, Gauteng',  10.0,  'Run',  NULL, 1),
  ('Cape Town Cycle Tour',   'Scenic cycling event around the Cape Peninsula.',
'2026-03-08 06:30:00', 'Cape Town, Western Cape', 109.0, 'Cycle', NULL, 2),
  ('Durban Beachfront Walk',  'Family-friendly charity walk along the Durban
beachfront.',      '2026-09-20 08:00:00', 'Durban, KwaZulu-Natal',  3.0,  'Walk',  NULL, 1);

-- Categories
INSERT INTO Categories (EventID, Name, DistanceKm, MaxParticipants)
VALUES
  (1, '5km Fun Run',  5.0,  500),
  (1, '10km Road Race',10.0, 300),
```

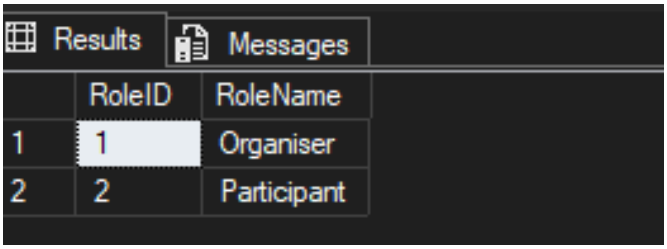
```
   (2, '109km Cycle',   109.0,1000),
   (2, '55km Cycle',    55.0, 800),
   (3, '3km Walk',      3.0,  400),
   (3, 'Under 20',      NULL, 150);

-- Enrolments
INSERT INTO Enrolments (UserID, CategoryID, Status)
VALUES
   (3, 1, 'Confirmed'), -- Sipho -> 5km Fun Run (Joburg)
   (4, 2, 'Confirmed'), -- Anika -> 10km Road Race (Joburg)
   (5, 3, 'Confirmed'), -- Naledi -> 109km Cycle (Cape Town)
   (3, 5, 'Confirmed'); -- Sipho -> 3km Walk (Durban)

-- Results for two of the enrolments
INSERT INTO Results (EnrolmentID, FinishTime, Position, Status)
VALUES
   (1, '00:24:12', 5,  'Finished'),
   (2, '00:52:47', 12, 'Finished');

PRINT 'RaceDay database schema created and seeded successfully.';

SELECT * FROM Roles;
SELECT * FROM Users;
SELECT * FROM Events;
SELECT * FROM Categories;
SELECT * FROM Enrolments;
SELECT * FROM Results;
```



Results | Messages

	UserID	FullName	Email	PasswordHash	RoleID	ProfilePictureURL	CreatedAt
1	1	Thabo Nkosi	thabo.nkosi@raceday.co.za	hashed_pw_1	1	NULL	2026-09-04 13:22:53.160
2	2	Lindiwe Dube	lindiwe.dube@raceday.co.za	hashed_pw_2	1	NULL	2026-09-04 13:22:53.160
3	3	Sipho Mokoena	sipho.mokoena@example.com	hashed_pw_3	2	NULL	2026-09-04 13:22:53.160
4	4	Anika van Wyk	anika.vanwyk@example.com	hashed_pw_4	2	NULL	2026-09-04 13:22:53.160
5	5	Naledi Molefe	naledi.molefe@example.com	hashed_pw_5	2	NULL	2026-09-04 13:22:53.160

Results | Messages

	EventID	Name	Description	EventDate	Location	DistanceKm	EventType	BannerImageURL	OrganiserID	CreatedAt
1	1	Johannesburg City Run	Annual road running event through the Johannesbu...	2026-11-14 07:00:00.000	Johannesburg, Gauteng	10.00	Run	NULL	1	2026-09-04 13:22:58.663
2	2	Cape Town Cycle Tour	Scenic cycling event around the Cape Peninsula.	2026-03-08 06:30:00.000	Cape Town, Western Cape	109.00	Cycle	NULL	2	2026-09-04 13:22:58.663
3	3	Durban Beachfront Walk	Family-friendly charity walk along the Durban beach...	2026-09-20 08:00:00.000	Durban, KwaZulu-Natal	3.00	Walk	NULL	1	2026-09-04 13:22:58.663

Results | Messages

	CategoryID	EventID	Name	DistanceKm	MaxParticipants
1	1	1	5km Fun Run	5.00	500
2	2	1	10km Road Race	10.00	300
3	3	2	109km Cycle	109.00	1000
4	4	2	55km Cycle	55.00	800
5	5	3	3km Walk	3.00	400
6	6	3	Under 20	NULL	150

Results | Messages

	EnrolmentID	UserID	CategoryID	EnrolmentDate	Status
1	1	3	1	2026-09-04 13:23:07.747	Confirmed
2	2	4	2	2026-09-04 13:23:07.747	Confirmed
3	3	5	3	2026-09-04 13:23:07.747	Confirmed
4	4	3	5	2026-09-04 13:23:07.747	Confirmed

Results | Messages

	ResultID	EnrolmentID	FinishTime	Position	Status
1	1	1	00:24:12.0000000	5	Finished
2	2	2	00:52:47.0000000	12	Finished

LINKS

Github: https://github.com/ST10493231/PROG6212-Part1.git
YouTube: https://youtu.be/sLPNnJ_g7q4?si=_8_nj2le3DNoH5yX

REFERENCES:

ERD: https://dbdiagram.io/d/RaceDay-ERD-6a9a9b3c5450bea1beed8bcc

ERD 2: RaceDay

CI Workflow: https://chatgpt.com/share/6a9ab3c6-16b4-83ea-a4ee-4b63ecdb92b4

Microsoft. (2026). Entity Framework Core documentation. Microsoft Learn. https://learn.microsoft.com/en-us/ef/core/

GitHub. (2026). GitHub Actions documentation. GitHub Docs. https://docs.github.com/en/actions